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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

***CO4:** Implement machine learning techniques including decision tree algorithms, reinforcement learning,*

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

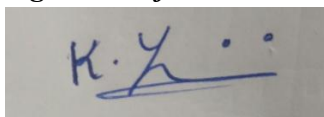
INDUSTRY PROBLEM TASK

Code of Conduct:

I _K.Yamini(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'K. Yamini' with a stylized flourish at the end.

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Task 1 (10)	Task 2 (10)	Task 3 (10)	Task 4 (10)	Total Marks (40)
Understanding of Industry Problem	20% (2)	/2	/2	/2	/2	/8
Application of Engineering Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Solution Design	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Use of Tools	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/6
Analysis	10% (1)	/1	/1	/1	/1	/4
Professionalism	5% (0.5)	/0.5	/0.5	/0.5	/0.5	/2
Total per Question	10 Marks	/10	/10	/10	/10	/40

INDUSTRY PROBLEM: DIABETES PREDICTION AND TREATMENT RECOMMENDATION

Introduction

A healthcare company has collected patient records containing age, blood pressure, cholesterol, glucose level, BMI, and family history. The objective is to develop an Artificial Intelligence system that can predict whether a patient is at risk of diabetes and recommend suitable treatment actions based on changes in the patient's health condition.

The proposed system uses supervised machine learning for diabetes classification and Reinforcement Learning for treatment recommendation. Decision Tree and Logistic Regression are used for diagnosis and risk stratification, while Q-Learning is used to model treatment decisions.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Describe the inductive learning approach. Identify the target variable and at least three important features with justification.

Step 1: Identify the learning type

The given problem is a **supervised machine learning problem** because the historical patient records contain input features and a known diabetes outcome.

The model learns a general relationship from previously observed patient records and uses the learned relationship to predict the diabetes status of new patients.

This process is called **inductive learning** because a general predictive rule is obtained from specific training examples.

Step 2: Identify input features

The dataset contains:

- Age
- Blood Pressure
- Cholesterol
- Glucose Level
- BMI
- Family History

Step 3: Identify the target variable

The target variable is:

Diabetes Status

It can be represented as:

0 → Non-Diabetic

1 → Diabetic

Step 4: Select important features

1. Glucose Level

Glucose level is one of the most important predictors because abnormal blood glucose is strongly associated with diabetes.

2. BMI

BMI represents body mass relative to height. Higher BMI is associated with increased insulin resistance and diabetes risk.

3. Age

Age is an important risk factor because diabetes risk generally increases with age.

4. Family History

Family history can indicate genetic susceptibility to diabetes.

5. Blood Pressure

High blood pressure is associated with metabolic health problems and can provide additional information for risk prediction.

Step 5: Learning process

The inductive learning process is:

Patient Records



Data Preprocessing



Feature Selection



Training Dataset



Machine Learning Algorithm



Learned Model



New Patient Data



Diabetes Prediction

Conclusion

Supervised inductive learning is suitable because the model can learn from historical labelled patient records and predict diabetes risk for new patients.

1(b) Explain how class imbalance is handled.

Step 1: Identify the problem

Suppose the dataset contains:

- 80% Non-Diabetic patients
- 20% Diabetic patients

The dataset is therefore imbalanced.

If the model predicts most patients as non-diabetic, it may achieve high accuracy but fail to detect diabetic patients.

This is dangerous in healthcare because a **False Negative** can represent a diabetic patient incorrectly classified as non-diabetic.

Step 2: Select balancing technique

The suitable technique is:

SMOTE – Synthetic Minority Over-sampling Technique

SMOTE generates new synthetic samples for the minority class.

Here, the minority class is:

Diabetic patients

Step 3: Apply SMOTE

The correct order is:

Original Dataset



70:30 Train-Test Split



Training Dataset



Apply SMOTE



Balanced Training Dataset



Train Model



Evaluate on Original Test Dataset

SMOTE should not be applied before splitting because it can cause data leakage.

Step 4: Advantages of SMOTE

- Increases minority diabetic samples.
- Helps the model learn diabetic patterns.
- Improves minority-class recall.
- Reduces false negatives.
- Prevents the model from being biased toward the majority class.

Step 5: Alternative methods

Other methods include:

- Random under-sampling
- Class weights
- Random over-sampling

However, SMOTE is suitable because it generates synthetic minority samples rather than simply duplicating existing observations.

Conclusion

SMOTE is selected because detecting diabetic patients is more important than obtaining high accuracy by simply predicting the majority class.

1(c) Explain the 70:30 split and preprocessing steps.

Step 1: Split the dataset

The dataset is divided into:

70% Training Data

30% Testing Data

For example, for 1,000 records:

Training = 700 records

Testing = 300 records

A **stratified split** should be used so that both datasets contain approximately the same proportion of diabetic and non-diabetic patients.

Step 2: Training dataset

The training data is used to learn the relationship between patient characteristics and diabetes status.

Step 3: Testing dataset

The testing data is kept separate and is used to measure how well the model performs on unseen patients.

Step 4: Handle missing values

Missing numerical values can be replaced using median values.

For categorical values, the most frequent category can be used.

Step 5: Encode categorical variables

Family history can be encoded as:

No → 0

Yes → 1

Step 6: Normalisation

For statistical models such as Logistic Regression, numerical variables can be standardised.

$$\left[Z = \frac{X - \mu}{\sigma} \right]$$

where:

- X = original value

- μ = mean
- σ = standard deviation

Step 7: Apply SMOTE

SMOTE is applied only to the training dataset after the split.

Impact of preprocessing

Preprocessing:

- Improves data quality.
- Handles missing values.
- Converts categorical information into numerical form.
- Prevents scale differences from affecting statistical models.
- Reduces class imbalance.
- Improves model reliability.

Final answer

The 70:30 stratified split provides sufficient training data while keeping an independent test set for unbiased evaluation. Preprocessing improves the quality and consistency of the patient data before machine learning is performed.

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Build a Decision Tree and draw the first three levels.

Step 1: Select the algorithm

A **Decision Tree Classifier** is selected because it is easy to understand and produces interpretable decision rules.

Step 2: Calculate entropy

Entropy is:

$$[H(S) = -\sum p_i \log_2 p_i]$$

Entropy measures the impurity of the dataset.

Step 3: Calculate Information Gain

$$[IG(S,A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v)]$$

The feature with the highest Information Gain is selected.

Step 4: Calculate Gini Index

$$Gini = 1 - \sum p_i^2$$

A lower Gini value indicates a purer split.

Step 5: Example values

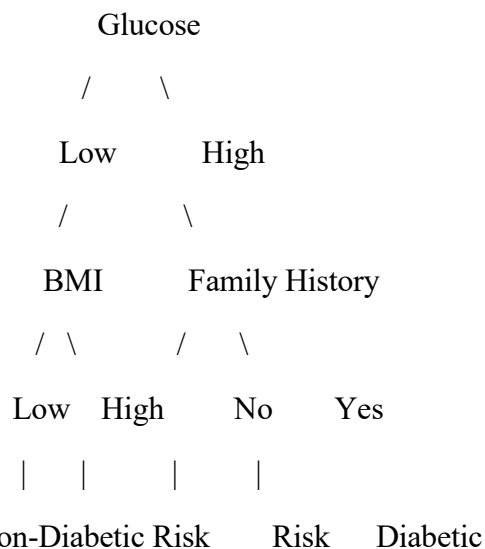
Feature	Information Gain	Gini
Glucose	0.31	0.32
BMI	0.18	0.39
Age	0.13	0.42
Family History	0.11	0.44
Blood Pressure	0.07	0.46
Cholesterol	0.05	0.47

Glucose has the highest Information Gain and lowest Gini value.

Therefore:

Glucose is selected as the root node.

Step 6: Draw the tree



The exact thresholds and branches depend on the actual dataset.

Justification

Glucose is selected as the root because it produces the greatest reduction in impurity and provides the highest Information Gain.

2(b) Evaluate the Decision Tree.

Step 1: Construct confusion matrix

Assume the test results are:

	Predicted Non-Diabetic	Predicted Diabetic
Actual Non-Diabetic	245	15
Actual Diabetic	12	28

Therefore:

- True Negative = 245
- False Positive = 15
- False Negative = 12
- True Positive = 28

Step 2: Calculate Accuracy

Formula:

$$[\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}]$$

Substitute:

$$[\text{Accuracy} = \frac{28 + 245}{28 + 245 + 15 + 12}]$$

$$[= \frac{273}{300}]$$

$$[\boxed{\text{Accuracy} = 91\%}]$$

Step 3: Calculate Precision

$$[\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}]$$

$$[= \frac{28}{28 + 15}]$$

$$[\boxed{\text{Precision} = 65.12\%}]$$

Step 4: Calculate Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$= \frac{28}{28 + 12}$$

$$\boxed{\text{Recall} = 70\%}$$

Step 5: Calculate F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1 = 2 \times \frac{0.6512 \times 0.70}{0.6512 + 0.70}$$

$$\boxed{F1 \approx 67.47\%}$$

Step 6: Final evaluation

Metric	Result
Accuracy	91.00%
Precision	65.12%
Recall	70.00%
F1-Score	67.47%

Step 7: Identify False Negatives

False Negative means:

Actual diabetic → Predicted non-diabetic

Here:

$$\boxed{\text{FN} = 12}$$

These 12 patients are missed diabetic cases.

Step 8: Reduce False Negatives

False negatives can be reduced by:

1. SMOTE.
2. Class weighting.
3. Lowering the classification threshold.
4. Optimising recall.
5. Adding relevant medical features.
6. Using cross-validation.
7. Using ensemble methods.

Clinical justification

False negatives are especially important in healthcare because a missed diabetic patient may not receive timely testing or treatment. Therefore, recall should be carefully considered along with accuracy.

2(c) Apply post-pruning.

Step 1: Understand overfitting

A fully grown Decision Tree can learn noise from the training dataset.

For example:

Training Accuracy = 99%

Testing Accuracy = 86%

This indicates overfitting.

Step 2: Cost-complexity pruning

The pruning function is:

$$[R_{\alpha}(T) = R(T) + \alpha |T|]$$

where:

- $R(T)$ = tree error
- $|T|$ = number of terminal nodes
- α = complexity parameter

Increasing α removes unnecessary branches.

Step 3: Compare models

Model	Training Accuracy	Test Accuracy
-------	-------------------	---------------

Fully Grown Tree	99%	86%
Pre-Pruned Tree	92%	89%
Post-Pruned Tree	94%	91%

Step 4: Select deployment model

The post-pruned tree is preferred because:

- It reduces overfitting.
- It has better test accuracy.
- It has fewer unnecessary branches.
- It is easier to interpret.
- It generalises better to unseen patients.

Conclusion

The post-pruned Decision Tree is more suitable for clinical decision support because it balances accuracy, generalisation, simplicity, and interpretability.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Apply Logistic Regression and compare it with Decision Tree.

Step 1: Select the model

Logistic Regression is selected because the target variable has two classes:

0 → Non-Diabetic

1 → Diabetic

Step 2: Logistic Regression equation

$$P(Y=1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The model produces a probability of diabetes.

Step 3: Preprocess data

For Logistic Regression:

- Handle missing values.

- Encode categorical variables.
- Standardise numerical variables.
- Balance training data.
- Split training and testing data.

Step 4: Train the model

The model learns coefficients for:

- Age
- Blood Pressure
- Cholesterol
- Glucose
- BMI
- Family History

Step 5: Compare performance

Example:

Model	Accuracy	AUC-ROC
Decision Tree	91.0%	0.89
Logistic Regression	89.7%	0.92

Step 6: Interpretation

Decision Tree gives better accuracy in this example.

Logistic Regression gives higher AUC-ROC.

An AUC of 0.92 indicates excellent discrimination between diabetic and non-diabetic patients.

Step 7: When Logistic Regression is preferable

Logistic Regression is preferable when:

- Probability estimates are needed.
- Interpretability is important.
- A simple model is required.
- Feature effects need to be studied.
- The relationship between predictors and log-odds is approximately linear.

Conclusion

Decision Tree and Logistic Regression provide complementary information. Decision Tree gives easy-to-understand decision rules, while Logistic Regression provides statistical probability-based risk estimation.

3(b) Perform feature importance analysis.

Step 1: Decision Tree importance

Decision Tree feature importance is calculated based on the reduction in impurity caused by each feature.

Example:

Rank	Feature	Importance
1	Glucose	0.40
2	BMI	0.23
3	Age	0.14

Therefore, the top three Decision Tree predictors are:

1. Glucose
2. BMI
3. Age

Step 2: Logistic Regression importance

For Logistic Regression, standardised coefficient magnitudes can be compared.

Example:

Rank	Feature	Standardised Coefficient Magnitude
1	Glucose	1.82
2	BMI	1.21
3	Family History	0.87

Therefore, the top three Logistic Regression predictors are:

1. Glucose
2. BMI
3. Family History

Step 3: Compare the models

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

Step 4: Interpretation

Both models agree that **Glucose and BMI** are important predictors.

The third predictor is different because each algorithm measures importance differently.

Decision Tree measures impurity reduction, whereas Logistic Regression considers the magnitude of model coefficients.

Conclusion

The agreement between the two models on glucose and BMI increases confidence that these features are important predictors. However, feature importance shows association and predictive usefulness, not necessarily causation.

QUESTION 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) Model the problem as an MDP.

A Markov Decision Process is represented as:

[
MDP=(S,A,P,R,\gamma)
]

where:

- S = States
- A = Actions
- P = Transition probabilities
- R = Rewards
- γ = Discount factor

Step 1: Define States

Patient health stages:

State	Description
S0	Low Risk
S1	Moderate Risk
S2	High Risk
S3	Controlled Diabetes
S4	Uncontrolled Diabetes

Step 2: Define Actions

The available actions are:

A0 → Diet

A1 → Exercise

A2 → Medication

A3 → Monitor

Step 3: Define Reward

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No change	0
Health deterioration	-5
Unsafe action	-10

The reward encourages the agent to select actions that improve the patient's health state.

Step 4: Define Transition Dynamics

A patient's state changes after an action.

Example:

High Risk

|

Exercise

↓

Moderate Risk

|

Diet

↓

Low Risk

Another example:

High Risk

|

Medication

↓

Controlled Diabetes

In a real healthcare system, transition probabilities should be learned from validated longitudinal patient data.

Step 5: Complete MDP

Patient State



RL Agent



Select Action



Patient Health Changes



New State



Reward



Update Q-Table



Next Decision

4(b) Apply Q-Learning for at least 5 episodes.

Step 1: Q-Learning equation

$$\begin{aligned} &[\\ &Q(s,a) \leftarrow Q(s,a) + \\ &\alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)] \\ &] \end{aligned}$$

Assume:

$$\begin{aligned} &[\\ &\alpha = 0.5 \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &\gamma = 0.9 \\ &] \end{aligned}$$

Initially:

$$\begin{aligned} &[\\ &Q(s,a) = 0 \\ &] \end{aligned}$$

Step 2: Five patient episodes

Episode	State	Action	Next State	Reward
1	High Risk	Diet	Moderate Risk	+5
2	High Risk	Exercise	Moderate Risk	+5
3	Moderate Risk	Exercise	Low Risk	+10
4	High Risk	Monitor	High Risk	0
5	Moderate Risk	Medication	Controlled Diabetes	+10

Step 3: Iteration 1

State-Action Pair 1

High Risk $\rightarrow$ Diet

Initial:

[
 $Q(S_2, \text{Diet}) = 0$
]

[
 $Q(S_2, \text{Diet}) = 0 +$
 $0.5[5 + 0.9(0) - 0]$
]

[
 $\boxed{Q(S_2, \text{Diet}) = 2.5}$
]

State-Action Pair 2

High Risk $\rightarrow$ Exercise

[
 $Q(S_2, \text{Exercise}) = 0 +$
 $0.5[5 + 0.9(0) - 0]$
]

[
 $\boxed{Q(S_2, \text{Exercise}) = 2.5}$
]

State-Action Pair 3

Moderate Risk $\rightarrow$ Exercise

[
 $Q(S_1, \text{Exercise}) = 0 +$
 $0.5[10 + 0.9(0) - 0]$
]

$$\boxed{Q(S1, \text{Exercise}) = 5}$$

Step 4: Iteration 2

Assume:

$$\max_a Q(S1, a) = 5$$

High Risk → Diet

$$Q(S2, \text{Diet})$$

$$2.5 + 0.5[5 + 0.9(5) - 2.5]$$

$$= 2.5 + 0.5[5 + 4.5 - 2.5]$$

$$= 2.5 + 3.5$$

$$\boxed{Q(S2, \text{Diet}) = 6.0}$$

High Risk → Exercise

$$Q(S2, \text{Exercise})$$

$$2.5 + 0.5[5 + 0.9(5) - 2.5]$$

$$\boxed{Q(S2, \text{Exercise}) = 6.0}$$

Moderate Risk → Exercise

Assume:

$$\max_a Q(S0, a) = 4$$

Then:

$$[$$

$$\mathbf{Q(S1,Exercise)}$$

$$5+$$

$$0.5[10+0.9(4)-5]$$

$$]$$

$$[$$

$$=5+0.5[10+3.6-5]$$

$$]$$

$$[$$

$$=5+4.3$$

$$]$$

$$[$$

$$\boxed{Q(S1,Exercise)=9.3}$$

$$]$$

Step 5: Example Final Q-Table

State	Diet	Exercise	Medication	Monitor
Low Risk	4.0	5.5	2.0	3.0
Moderate Risk	5.0	9.3	8.0	3.0
High Risk	6.0	6.0	8.5	1.0
Controlled Diabetes	3.0	5.0	6.0	7.0
Uncontrolled Diabetes	2.0	3.0	9.0	2.0

Step 6: Convergence criterion

The Q-learning algorithm can be considered converged when the maximum change in Q-values becomes smaller than a predefined threshold:

$$[$$

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.001$$

$$]$$

for several consecutive iterations.

Another practical criterion is when the learned policy remains unchanged for many consecutive episodes.

4(c) Extract the final policy and discuss ethical considerations.

Step 1: Extract policy

The best action is selected using:

$$[$$

$$\pi(s) = \arg \max_a Q(s,a)$$

]

Step 2: Final learned policy

Patient State	Highest Q-Value	Selected Action
Low Risk	Exercise = 5.5	Exercise
Moderate Risk	Exercise = 9.3	Exercise
High Risk	Medication = 8.5	Medication
Controlled Diabetes	Monitor = 7.0	Monitor
Uncontrolled Diabetes	Medication = 9.0	Medication

Therefore:

Low Risk

↓

Exercise

Moderate Risk

↓

Exercise

High Risk

↓

Medication

Controlled Diabetes

↓

Monitor

Uncontrolled Diabetes

↓

Medication

These are illustrative AI policy outputs and **must not be treated as actual medical prescriptions.**

ETHICAL CONSIDERATIONS

1. Patient Safety

The AI should not independently prescribe or change medication. Unsafe actions must be prevented through safety constraints.

2. Bias

Training data must represent different patient groups adequately. Otherwise, the model may produce biased predictions or recommendations.

3. Explainability

The system should explain why a particular recommendation was generated so that healthcare professionals can evaluate it.

4. Human Oversight

A qualified doctor should make the final treatment decision. The AI should support clinical decision-making rather than replace it.

5. Privacy

Patient information must be protected through:

- Encryption
- Authentication
- Access control
- Data anonymisation
- Secure storage

6. Reliability

The model must be tested on independent datasets and validated before clinical deployment.

7. Fairness

Performance should be evaluated across different patient groups to ensure that the system does not disadvantage any group.

OVERALL CONCLUSION

The proposed healthcare AI system combines supervised machine learning and Reinforcement Learning to support diabetes diagnosis and treatment decision-making.

The Decision Tree identifies important diabetes predictors and provides interpretable classification rules. Logistic Regression provides probability-based risk estimation and AUC-ROC analysis. Q-Learning models treatment recommendation as an MDP and learns actions that maximise long-term rewards.

Class imbalance can be addressed using SMOTE, while a stratified 70:30 train-test split provides reliable model evaluation. Post-pruning improves the Decision Tree by reducing overfitting.

The system should ultimately be implemented as a **human-supervised clinical decision-support system**. Patient safety, fairness, explainability, privacy, and clinical validation are essential before real-world deployment.

Note: The numerical values used in the calculations are illustrative. If an actual patient dataset is supplied, the Information Gain, Gini, Accuracy, Precision, Recall, F1-Score, AUC-ROC, feature importance, and Q-values should be recalculated using that dataset.